📓 SOC Internship – Day 5 Notes

Topic: Introduction to Logs & Log Analysis

🎯 Objective

Understand what logs are and how SOC analysts use them to detect cyber attacks.

📃 1. What are Logs?

Logs are records of events that occur in a system.

Examples:

- Login attempts
- File access
- System errors

👉 In simple terms:
Logs = **activity history of a system**

🔍 2. Why Logs are Important

Logs play a critical role in cybersecurity:

- Detect cyber attacks
- Track user activity
- Investigate security incidents

👉 SOC analysts mainly depend on logs to monitor threats.

🗂 3. Types of Logs

1. System Logs

- OS-related events
- Example: system startup, shutdown

2. Application Logs

- Events from applications
- Example: browser or software errors

3. Security Logs

- Authentication & security events
- Example: login success/failure

🕵 4. Example Log Entry

Failed login from IP 192.168.1.10 at 10:30 PM

👉 Shows:

- Action: Failed login
- Source: IP address
- Time: When it happened

🧪 5. Log Analysis Activity

📌 Given Logs:

Apr 10 10:17:45 Failed password for user1 from 192.168.1.10
Apr 10 10:18:02 Failed password for user1 from 192.168.1.10
Apr 10 10:18:15 Failed password for user1 from 192.168.1.10
Apr 10 10:18:30 Failed password for user1 from 192.168.1.10

Apr 10 11:05:10 Accepted password for admin from 203.0.113.45
Apr 10 11:06:22 Accepted password for admin from 203.0.113.45

Apr 10 12:12:11 Failed password for user2 from 192.168.1.20
Apr 10 12:12:15 Failed password for user2 from 192.168.1.20
Apr 10 12:15:30 Accepted password for user2 from 192.168.1.20

❓ Answers to Activity

1. Which IP looks suspicious?

👉 **192.168.1.10**

2. What type of attack is this?

👉 **Brute Force Attack**

3. Why is it suspicious?

- Multiple failed login attempts in a short time
- Same IP repeatedly trying passwords
- Indicates automated attack attempt

🧠 Key Takeaways

- Logs are essential for **detecting threats**
- Repeated failures = possible **attack pattern**
- SOC analysts must **monitor & analyze logs continuously**